

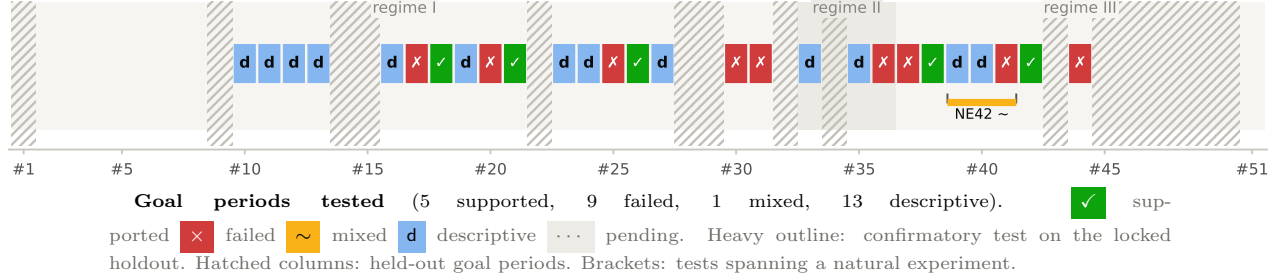
H31: Consensus time scales with the interaction graph's spectral gap

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2.5/5

Bottom line. Consensus in the village does not follow the $1/\lambda_2$ law: gradual project consensus is forecast no better than by a constant (≈ 4.5 active h) in attention and in work, half of all consensus is set by the kickoff or arrives in one-window waves, and deadline-bound votes settle at the speed of read-out.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Consensus events were defined by rule before outcomes and timed across non-holdout periods #10–#44: project-share majorities (63 events, 20 periods), #26 votes, and content convergence. Their times were regressed on the exposure graph's spectral gap λ_2 . Gradual project consensus (median 4.5 active h) scales sub-linearly, $\tau \propto \lambda_2^{-0.38}$ [0.05, 0.60], not as $1/\lambda_2$; the slope-1 law loses to a constant out of sample (synthetic power was 0.13). About half of all consensus is frozen at the kickoff or arrives in one-window waves. Content never converged. **Round 1b** (context-ledger visibility, deterministic labels, DQ4 work ledger, DQ6 ballots) changes no card verdict, weakens the sub-linear lead (CI touches 0 under controls), and shows work commits reaching majorities on the same repos at least as fast (2.0 vs 4.6 h). On the correct #26 rounds the votes settle in 57 s and 39 s: two to three read-out cycles, 70–100× faster than the λ_2 forecast.

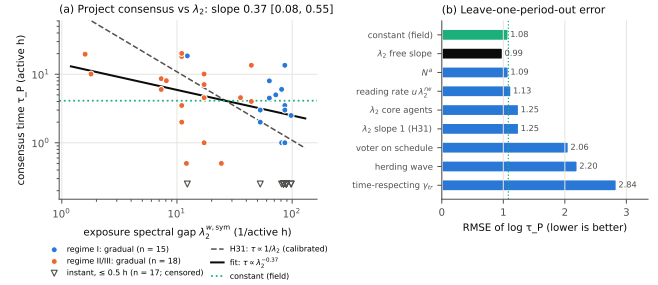
PHYSICAL MODEL

Model 10's voter and Potts coarsening, plus graph diffusion (HH68). W_{ij} : agent j 's messages first seen by i (round 1b: context-ledger visibility), per active hour. L is its symmetrized Laplacian, with spectral gap λ_2 . The state x_i is an agent's project (shared labels from artifact mentions, or work commits) or its whitened statement embedding. Rivals run on the same reading schedule: DeGroot averaging (time-respecting gap γ_{tr}), the voter model ($\tau \propto N/u$, u = reading rate), a herding wave, and a field (kick-locked timing).

$$L = D - \frac{1}{2}(W + W^T), \quad 0 = \mu_1 < \lambda_2 \leq \mu_3 \leq \dots$$

$$\dot{x}_i = \sum_j W_{ij} (x_j - x_i) \Rightarrow \tau \propto 1/\lambda_2 \quad (\text{H31}), \quad \tau_{\text{voter}} \propto N/u$$

$$\log \tau_e = c + b \log(1/\lambda_2) + \epsilon_e : \quad \hat{b} = 0.38 [0.05, 0.60] \quad \text{vs. } b = 1$$



Observables. (a) Gradual project-consensus times against the exposure spectral gap, with the calibrated slope-1 law (dashed), the free fit (solid) and the constant forecast (dotted). Triangles: instant one-window events ($\tau \leq 0.5$ h, censored). (b) Leave-one-period-out error of each forecast rule (round 1; round 1b moves every number by ≤ 0.1 and lets the reading-rate predictor $u\lambda_2^{rw}$ edge the constant by 2%). The voter, herding-wave and time-respecting models miss by an order of magnitude.

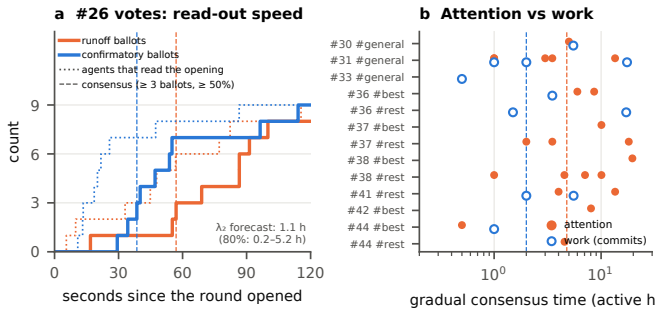
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly negative, with two usable numbers. The textbook “consensus time $\propto 1/\lambda_2$ ” is no use to a swarm operator: in broadcast rooms λ_2 is mostly message volume, and consensus comes in three kinds (kickoff-set, one-window waves, hours-long). For a project, expect a majority about 4.5 active hours after it first appears (80%: 1–13.5 h), and expect the work commits to get there as fast as the chatter. For a deadline-bound vote, expect consensus within two or three read-out cycles of the call to vote: tens of seconds, not hours. So size vote windows to read-out delays, not to the social graph. Content alignment never converged; it decayed after the kickoff. More talk did not buy agreement in content.

H31-consensus-time-spectral-gap · generated 2026-10-04 11:10 by infra/summaries/build_summaries.py · card hypotheses/H31-consensus-time-spectral-gap/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0)

RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b)	Verdict
P1 slope 1; beats constant	0.37 → 0.38 [0.05, 0.60]; RMSE 1.24 vs 1.08	failed
P3 kick-locked	0.41 vs 0.25, all kickoff-frozen	trivial
P4 $\geq 60\%$ abrupt	13% → 17%	failed
P5 content slope	no convergence (12 diverge)	untestable
P6 room sign as D	3/4 → 2/3	weak
P8 power ≥ 0.5	0.13	failed
P10 vote ≤ 1 h	12.5 h (wrong vote) → 57 s runoff, 39 s confirmatory	failed → supported
R1b-2 work slower, fewer frozen	19 vs 23 events; 2.0 vs 4.6 h; frozen 37% vs 39%	mostly failed
#26 $\tau_V \approx \tau_{read}(3)$	1.7×, 2.9×	partly
NE42 λ_2 A-B-A	up 2.4× at merge, not down at split	failed



Second figure. (a) #26 on DQ6 ballots: ballots (solid) track the agents that have read the round's opening (dotted); consensus (dashed) comes 57 s and 39 s after the opening. (b) Gradual consensus times per block from #30 on, attention (filled) vs work commits (open): work is not slower (medians dashed).

GOAL PERIODS IN DETAIL

G10 d 1b: descriptive (unchanged)
 G11 d 1b: descriptive (unchanged)
 G12 d 1b: descriptive (unchanged)
 G13 d 1b: descriptive (unchanged)
 G16 d 1b: descriptive (unchanged)
 G17 × 1b: failed (unchanged)
 G18 ✓ 1b: supported (unchanged)
 G19 d 1b: descriptive (was supported; label renumbering)
 G20 × 1b: failed (unchanged)
 G21 ✓ 1b: supported (was failed; label renumbering)
 +18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Thirty-one gradual events from 13 periods and one primary test, which failed. The sub-linear slope ($p = 0.07$) loses significance under a regime effect, at the block level and without the pre-NE09 periods. λ_2 , message volume, N and regime co-vary. The frozen/instant/gradual split and the structure decomposition are post hoc. The leave-one-period-out winner flips between the constant and $u\lambda_2^{rw}$ on 2% margins; the frozen holdout rule is unchanged. #26 has three rounds and NE42 is goal-confounded and not reversed. Work consensus uses public commits (dense from #30). Content uses one embedding model.

ROUND 2

Holdout run (#29, #46, #47, #49, #50) after sign-off. An agent-level test of uneven exposure (do poorly connected agents adopt later, in attention and in commits?). Read-out-gated vote model for any deadline-bound call (H53's clock), checked on #12 verdicts.

No round-2 redirects yet.